CANTERBURY BANKSTOWN

3D Digital Model Submission Manual

Requirements to submit a 3D model for development applications and state significant development applications.

Introduction

The purpose of this manual is to provide technical guidance for the submission of 3D digital models of development applications to the City of Canterbury Bankstown.

The submitted 3D models will be inserted into Council's local government area-wide 3D digital model, to keep it up to date. They will also be used for detailed development assessment and to inform the community about new development applications.

From **1 July 2026**, Council is requiring 3D models to be submitted with larger development applications (applicable typologies can be found below). Prior to this date submitting a 3D model is optional but encouraged.

When is a 3D model submission required?

A 3D model is required to be submitted with all state significant development applications, development applications, amended plans or Section 4.55 applications that propose a new building or modifications to an existing building that is three or more storeys and of the following typologies:

Residential flat buildings, shop top housing, multi dwelling housing, boarding houses, seniors housing, co-living housing, commercial premises, schools, warehouse or distribution centres, places of worship, centre-based childcare facilities and registered clubs.

Any amendment to the design at the development application stage will require a resubmission of a 3D model. 3D models may also be requested for Land and Environment Court cases, design review panels or as a result of council meetings or negotiated options.

An 'as built' model should also be submitted before an occupation certificate is issued and this will form a condition of consent.

Technical specifications

Level of detail

All buildings within proposed site boundaries need to be modelled in detail, including existing buildings that will be retained or adapted in the development.

- All external geometry must be modelled accurately. This includes the accurate placement of glazing, balconies, roof pitches, terraces, roof services and any other external feature such as entrances, fencing or services.
- Only the external building envelope is to be provided. Unnecessary features such as internal layouts, stairs, walls, furniture or building maintenance equipment must be deleted as they amount to large file sizes.
- Complex external vegetation features are also discouraged, but landscape/deep soil areas should be outlined in colour or texture. Elements such as benches, lighting and barbeques should be included but in simple geometry.
- Ensure that there are no missing or displaced elements of the model after exporting.

For residential areas, neighbouring dwellings likely impacted by the proposal may need to be included as part of the submission. Liaise with Council's Digital Urban Designer to understand if this is required.

For all other areas, Council may provide existing and desirable future character neighbouring buildings. Liaise with Council's Digital Urban Designer to request a copy.

File formats

- 3D digital models should be provided to the City of Canterbury Bankstown in one of the following formats: skp, 3ds, collada, obj, 3dm.
- Proposed buildings, terrain and context models (as required) must be submitted in separate layers or files.

Units

- The base units should be in metres and at a scale of 1:1.
- When not georeferenced, models should have both the origin and pivot point at 0,0,0 and rotated in accordance with the cadastral alignment.
- A plan that indicates applicable setbacks from the cadastral boundary should also be provided if the model does not occupy the entire site, for the purposes of accurate placement.

Textures

Models should be submitted with either textures or colours.

- Textures should be in JPG, TIFF, PNG or TGA format and, whenever possible, should be kept to a minimum, with an ideal texture size being no larger than 512 x 512 pixels.
- Texture names should only contain A-Z and 0-9 characters.

Models should be purged of all unused texture and object links.

File naming conventions

Models should be named according to the file export date, stage/type of the application ('PP'), and site address:

YYYYMMDD_ApplicationStage_Address_Suburb.fbx

For example:

20240129_PP_1NorthTerrace_Bankstown.fbx

How to submit the 3D model?

3D models should be submitted alongside with other applicable documentation when lodging a development application on the Planning Portal. Since the portal does not accept 3D model formats, models need to be compressed into a .zip file.

In case there are any issues uploading the model, please send them directly to Council's Digital Urban Designer.

Further information

For any technical questions, please contact:

3dmodel@cbcity.nsw.gov.au